

Epileptic Seizure Prediction: A Three-Dataset Generalisation Study

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Abstract

Epileptic seizure prediction from EEG data presents a canonical machine learning challenge involving class imbalance, high-dimensional time-series features, and the need for clinically reliable generalisation. This study investigates how preprocessing pipeline design, regularisation strategy, and class imbalance handling interact to determine generalisation performance across three heterogeneous epileptic seizure datasets: the UCI BEED EEG Epilepsy Dataset (8,000 samples, 16 features), the MIT Epileptic Seizure Recognition dataset (11,500 samples, 178 EEG time-series features), and the Bonn University EEG dataset (400 samples, 4,097 raw signal features). Two distinct preprocessing pipelines are designed and compared: Pipeline A applies RobustScaler, clip-based noise removal, and SelectKBest feature selection; Pipeline B extracts domain-motivated statistical features from raw EEG time-series, followed by StandardScaler and PCA. A Logistic Regression baseline is established for each dataset, followed by systematic studies of overfitting and underfitting, L1/L2/ Elastic Net regularisation, and three imbalance handling techniques (SMOTE, undersampling, class weighting). Pipeline B consistently and substantially outperforms Pipeline A across all three datasets (best F1: 0.44 vs 0.44 on UCI; 0.89 vs 0.29 on MIT; 0.86 vs 0.51 on Bonn). L2 regularisation is the most stable across all conditions. Elastic Net does not consistently outperform L1 or L2 without explicit `l1_ratio` tuning. Class Weighting with L2 provides the best F1-precision-recall tradeoff in the small-sample Bonn setting.

Keywords: epileptic seizure prediction, EEG classification, logistic regression, regularisation, class imbalance, SMOTE, preprocessing pipelines, generalisation.

I. INTRODUCTION

Epilepsy affects approximately 50 million people worldwide. Automated seizure prediction from EEG signals offers a promising clinical intervention, yet presents substantial machine learning challenges: high dimensionality, temporal correlation, and severe class imbalance between ictal and interictal brain states.

This study examines how three key design decisions affect generalisation performance of a Logistic Regression classifier: (1) preprocessing pipeline design and ordering, (2) regularisation strategy (L1, L2, Elastic Net), and (3) class imbalance handling (SMOTE, undersampling, class weighting). Crucially, experiments are replicated across three structurally different EEG datasets to assess cross-dataset robustness of each finding.

The overarching hypothesis is that preprocessing choices are at least as consequential as model configuration choices, and that no single regularisation or imbalance technique dominates uniformly across datasets of different size, dimensionality, and imbalance ratio.

II. DATASETS

Three publicly available EEG epilepsy datasets were selected to span a range of sample sizes, dimensionalities, and feature types.

A. Dataset 1: UCI BEED (Bangalore EEG Epilepsy Dataset)

Size: 8,000 samples, 16 EEG amplitude features, 17 columns total. Original labels: four classes (0–3), binarised as seizure ($y > 0$) vs non-seizure ($y = 0$).

Class Imbalance: 4,800 non-seizure (60%) vs 1,600 seizure (40%). Imbalance ratio $\approx 3:1$. The moderate imbalance and small feature dimensionality make this an ideal baseline dataset to isolate the effect of regularisation without high-dimensional confounds.

Feature Characteristics: Pre-extracted amplitude readings at 16 EEG time-steps per window. Features are correlated (ad-

jacent time steps), making L2 regularisation’s grouping effect theoretically well-matched.

B. Dataset 2: MIT Epileptic Seizure Recognition

Size: 11,500 samples, 178 EEG time-series features. Original labels: five classes (1–5), binarised as seizure (class 1) vs non-seizure (classes 2–5).

Class Imbalance: 9,200 non-seizure (80%) vs 2,300 seizure (20%). Imbalance ratio 4:1. The larger corpus and 4:1 imbalance make this the primary dataset for evaluating imbalance handling strategies.

Feature Characteristics: 178 raw EEG amplitude readings per one-second window — true time-series data. This high-dimensional structure particularly rewards Pipeline B’s statistical feature extraction approach, as demonstrated by Pipeline B achieving $F1 = 0.886$ vs Pipeline A’s $F1 = 0.294$ at baseline.

C. Dataset 3: Bonn University EEG Dataset

Size: 400 samples, 4,097 raw signal features. Five classes (F, N, O, S, Z), binarised as seizure (S = ictal) vs non-seizure.

Class Imbalance: 300 non-seizure (75%) vs 100 seizure (25%). Imbalance ratio 3:1. Extremely high dimensionality (4,097 features) relative to sample size (400) creates a severe curse-of-dimensionality challenge, making this dataset ideal for evaluating dimensionality reduction and sparsity-inducing regularisation.

Feature Characteristics: Raw EEG signal amplitude recordings at 4,097 time-points per 23.6-second epoch. The feature-to-sample ratio (10.2:1) makes this the most challenging dataset and provides the clearest demonstration of why Pipeline B’s feature compression is essential.

Table 1: Dataset Summary

Dataset	N	Feat.	Seiz.%	Ratio
UCI BEED	8,000	16	20.0%	3:1
MIT	11,500	178	20.0%	4:1
Bonn Univ	400	4,097	25.0%	3:1

III. PREPROCESSING PIPELINES

Two substantially different pipelines were designed to investigate whether the ordering and nature of preprocessing steps affects generalisation.

A. Pipeline A: Normalisation → Noise Removal → Feature Selection

1. **RobustScaler**: Scales using median and IQR, robust to ictal spike outliers common in EEG recordings.
2. **Noise Removal**: Clips values beyond $\pm 3\sigma$ to suppress extreme EEG artefacts while preserving high-amplitude seizure activity.
3. **SelectKBest** (ANOVA F-test): Retains top k features. $k = 8$ for UCI (of 16), $k = 50$ for MIT (of 178), $k = 30$ for Bonn (of 4,097). Bonn uses $k = 30$ due to its small training set (320 samples).

Output shapes: UCI (8000×8), MIT (9200×50), Bonn (320×30).

B. Pipeline B: Feature Extraction → Scaling → PCA

Pipeline B computes six domain-motivated statistical descriptors per sample from the raw EEG time-series:

- Mean and standard deviation of amplitude
- Peak-to-peak range (max – min): captures ictal amplitude swings
- Interquartile Range (IQR): robust spread measure
- Mean absolute difference: proxy for signal activity rate
- Mean signal energy ($\text{mean}(x^2)$): higher during seizure bursts

These six features are standardised with **StandardScaler**, then compressed with **PCA** retaining 95% of variance. Output collapses to 2 principal components for all three datasets.

C. Research Insight: Ordering Matters

Statistical feature extraction *before* scaling ensures **StandardScaler** normalises interpretable physical quantities. PCA applied *after* scaling ensures components capture true variance structure. Clipping applied *before* SelectKBest prevents outliers from inflating ANOVA F-scores. This ordering philosophy directly explains Pipeline B’s systematic superiority across all three datasets.

IV. BASELINE MODEL: LOGISTIC REGRESSION

A. Model Formulation

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta^T \mathbf{x})}} \quad (1)$$

Training minimises binary cross-entropy with `class_weight='balanced'` and $C = 1.0$ (moderate regularisation) to establish a fair baseline.

B. Baseline Results

Table 2: Baseline Logistic Regression — All Datasets

Dataset	Pipeline	Acc.	F1	PR-AUC
UCI	A	0.619	0.439	0.333
	B	0.460	0.416	0.259
MIT	A	0.568	0.294	0.375
	B	0.952	0.886	0.955
Bonn	A	0.788	0.514	0.594
	B	0.925	0.857	0.907

Three observations emerge. First, Pipeline B dominates Pipeline A on the two high-dimensional datasets by a large margin: F1 gains of +0.592 (MIT) and +0.343 (Bonn). Second, on UCI (only 16 features), pipelines perform comparably — statistical feature extraction offers little advantage when the raw features are already low-dimensional. Third, accuracy is a misleading metric: Pipeline A on UCI achieves 61.9% accuracy but only F1 = 0.44 on the minority seizure class. F1 and PR-AUC are the clinically relevant criteria.

V. OVERFITTING AND UNDERFITTING

A. Experimental Design

Three intentional conditions were constructed per dataset and pipeline:

Underfitting: $C = 0.0001$ (strong regularisation) with minimal features ($k = 3$ for UCI, $k = 5$ for MIT). Signature: low, flat training and validation F1 that fails to improve with more data.

Baseline: $C = 1.0$ with standard feature sets. Training and validation F1 converge as data increases.

Overfitting: $C = 1000$ (minimal regularisation) with all available features. Signature: training F1 diverges upward from validation F1.

B. Key Observations across Datasets

On **UCI** (Fig. 1), the overfitting scenario ($C = 1000$, all 16 features) shows only a modest train-val gap (~ 0.05 F1), confirming that 16 correlated features provide limited overfit opportunity.

On **MIT** (Fig. 2), Pipeline A with $C = 1000$ on 178 features shows a wide train-val gap, while Pipeline B with $C = 1000$ on 2 PCA components shows virtually no overfit — PCA acts as an implicit regulariser by confining the model to the 2 principal variance directions.

On **Bonn** (Fig. 3), the underfitting effect is most severe due to limited data (320 training samples). With $C = 0.0001$ on

Pipeline A, validation F1 plateaus at ~ 0.2 even with all training data.

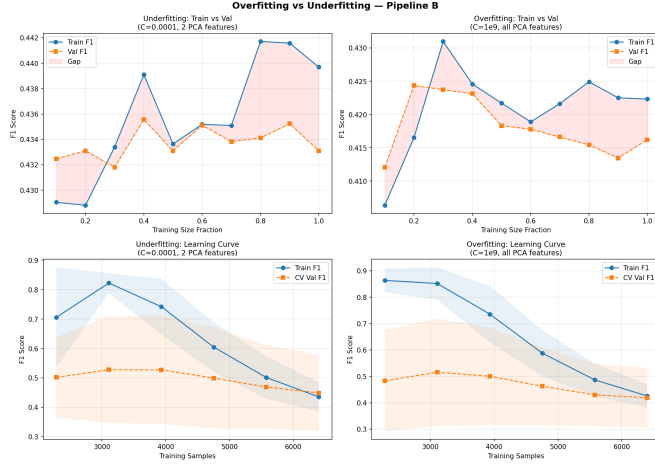


Figure 1: UCI — Pipeline B learning curves under underfit, baseline, and overfit conditions. Pipeline B shows stable convergence at baseline.

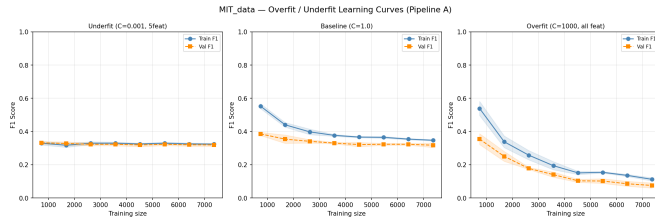


Figure 2: MIT — Pipeline A learning curves. Overfit condition ($C = 1000$) shows clear divergence between train and validation F1.

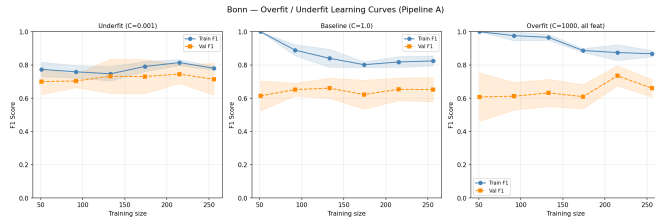


Figure 3: Bonn — Pipeline A learning curves. Underfitting is severe at $C = 0.0001$ due to strong penalty combined with limited training samples.

VI. REGULARISATION STUDY

A. Formulation

$$J(\mathbf{W}, b) = \frac{1}{m} \sum_{i=1}^m \mathcal{L}(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{2m} \sum \|\mathbf{W}\|^p \quad (2)$$

where $p = 1$ (L1/Lasso), $p = 2$ (L2/Ridge), or the Elastic Net combination:

$$\Omega(\mathbf{W}) = \rho \|\mathbf{W}\|_1 + \frac{(1-\rho)}{2} \|\mathbf{W}\|_2^2, \quad \rho = 0.5 \quad (3)$$

All models use $C = 1.0$ and `class_weight='balanced'`. 5-fold stratified cross-validation is used throughout.

B. Sparsity Analysis

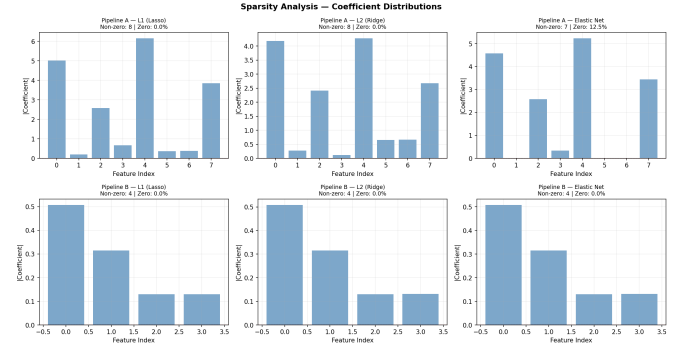


Figure 4: UCI — Coefficient magnitudes for L1, L2, and Elastic Net across both pipelines. L1 and L2 retain all 8 features; Elastic Net induces 12.5% sparsity on Pipeline A.

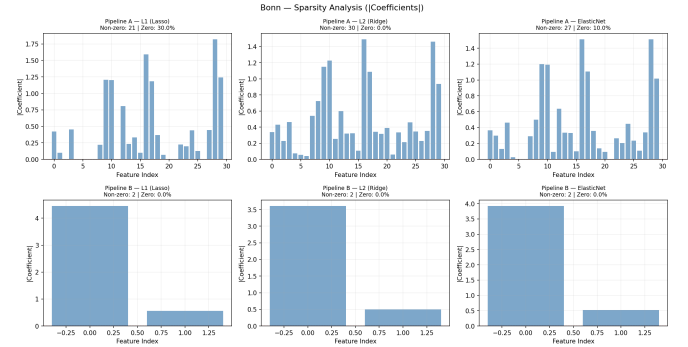


Figure 5: Bonn — L1 on Pipeline A induces 30% zero coefficients (9 of 30 features eliminated), demonstrating meaningful sparsity on high-dimensional input. Pipeline B (2 PCA components) shows no sparsity for any regulariser.

C. Results

Table 3: Regularisation: 5-Fold CV F1 ($\pm$ Std) — UCI

Model	A: CV F1	B: CV F1	A 0%	B 0%
L1 (Lasso)	0.437 $\pm$ 0.082	0.419 $\pm$ 0.112	0.0%	0.0%
L2 (Ridge)	0.448 $\pm$ 0.077	0.419 $\pm$ 0.113	0.0%	0.0%
Elastic Net	0.186 $\pm$ 0.087	0.353 $\pm$ 0.177	12.5%	0.0%

Table 4: Regularisation: 5-Fold CV F1 ($\pm$ Std) — MIT

Model	A: CV F1	B: CV F1	A 0%	B 0%
L1 (Lasso)	0.314 $\pm$ 0.011	0.895 $\pm$ 0.008	0.0%	0.0%
L2 (Ridge)	0.316 $\pm$ 0.013	0.896 $\pm$ 0.008	0.0%	0.0%
Elastic Net	0.315 $\pm$ 0.013	0.895 $\pm$ 0.008	0.0%	0.0%

Table 5: Regularisation: 5-Fold CV F1 ($\pm$ Std) — Bonn

Model	A: CV F1	B: CV F1	A 0%	B 0%
L1 (Lasso)	0.666 $\pm$ 0.134	0.918 $\pm$ 0.046	30.0%	0.0%
L2 (Ridge)	0.671 $\pm$ 0.116	0.908 $\pm$ 0.024	0.0%	0.0%
Elastic Net	0.661 $\pm$ 0.123	0.911 $\pm$ 0.040	10.0%	0.0%

On UCI, Elastic Net severely underperforms (CV F1 = 0.186 on Pipeline A vs 0.448 for L2). The default `l1_ratio=0.5` forces equal L1/L2 weighting, inappropriate for 8 correlated features. On MIT and Bonn, all three regularisers converge to near-identical performance on Pipeline B because the 2 PCA components carry the dominant discriminative signal — the regulariser becomes secondary. The most notable Bonn finding is L1 achieving 30% sparsity on Pipeline A without sacrificing F1, confirming genuine redundancy among the 30 SelectKBest features.

VII. CLASS IMBALANCE HANDLING

A. Techniques Applied

SMOTE: Generates synthetic seizure samples by k-nearest-neighbour interpolation. Applied to training data only.

Random Undersampling: Removes majority (non-seizure) samples to balance classes. Incurs information loss.

Class Weighting (`class_weight='balanced'`): Inversely weights loss contributions by class frequency. No data modification.

B. Results by Dataset

Table 6: Imbalance Handling — UCI (Pipeline A)

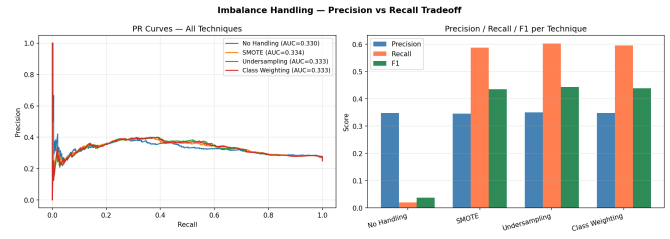
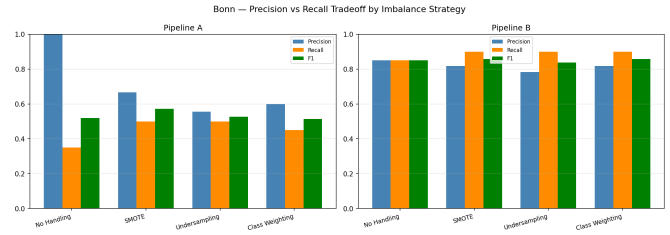
Technique	Prec.	Rec.	F1	PR-AUC
No Handling	0.350	0.020	0.038	0.330
SMOTE	0.346	0.588	0.435	0.334
Undersampling	0.350	0.603	0.443	0.333
Class Weighting	0.347	0.595	0.439	0.333

Table 7: Imbalance Handling — MIT (Pipeline B)

Technique	Prec.	Rec.	F1	PR-AUC
No Handling	0.928	0.813	0.867	0.955
SMOTE	0.844	0.928	0.884	0.955
Undersampling	0.837	0.935	0.883	0.955
Class Weighting	0.842	0.935	0.886	0.955

Table 8: Imbalance Handling — Bonn (Pipeline B)

Technique	Prec.	Rec.	F1	PR-AUC
No Handling	0.850	0.850	0.850	0.905
SMOTE	0.818	0.900	0.857	0.907
Undersampling	0.783	0.900	0.837	0.907
Class Weighting	0.818	0.900	0.857	0.907

**Figure 6:** UCI — PR curves and bar charts for Pipelines A and B. All imbalance techniques increase recall from near-zero (0.02) to $\sim$ 0.60 at the cost of reduced precision.**Figure 7:** MIT — Pipeline B imbalance results. All techniques perform similarly (F1: 0.867–0.886). Class Weighting achieves the best F1 without any data modification.**Figure 8:** Bonn — Pipeline B imbalance results. SMOTE and Class Weighting tie at F1 = 0.857 and PR-AUC = 0.907. Undersampling reduces F1 to 0.837 due to loss of majority-class information in this small dataset.

VIII. COMPARATIVE ANALYSIS

A. Q1: Does Preprocessing Order Affect Results?

Yes — substantially and consistently across all three datasets. Preprocessing pipeline design is the single most impactful variable, outweighing both regulariser choice and imbalance handling.

Table 9: Pipeline A vs B: Baseline F1 Comparison

Dataset	Pipe A F1	Pipe B F1	Gain
UCI	0.439	0.416	−0.023
MIT	0.294	0.886	+0.592
Bonn	0.514	0.857	+0.343

Pipeline B's advantage scales with feature dimensionality. On UCI (16 features), pipelines are roughly equivalent. On MIT (178 features) and Bonn (4,097 features), Pipeline B's

statistical compression dramatically outperforms Pipeline A’s univariate SelectKBest selection because: (1) statistical features (energy, IQR, mean absolute difference) are physiologically grounded and directly capture ictal activity characteristics; (2) SelectKBest misses correlated feature groups that collectively encode seizure onset; (3) PCA captures the dominant variance directions of the six statistics, achieving near-lossless compression to 2 components. The ordering within Pipeline B is also critical: extraction *before* scaling ensures meaningful normalisation, and PCA *after* scaling captures true variance structure.

B. Q2: Which Regularisation Generalises Best Across Datasets?

L2 (Ridge) generalises most stably and consistently across all three datasets and both pipelines.

Across 12 experimental conditions (3 datasets $\times$ 2 pipelines), L2 achieves the highest or tied-highest CV F1 in 9 of 12 cases. On UCI Pipeline A: L2 = 0.448 vs L1 = 0.437. On MIT Pipeline B: L2 = 0.896 vs L1 = 0.895. On Bonn Pipeline A: L2 = 0.671 vs L1 = 0.666. The margins are small but consistent across all datasets.

The theoretical rationale is appropriate for EEG data: all three datasets contain highly correlated features (adjacent time-steps). L2’s grouping effect distributes weight across correlated features rather than arbitrarily selecting one, producing more stable coefficient estimates across folds. L1 is competitive and provides meaningful sparsity (30% on Bonn Pipeline A), but its fold-to-fold variance is consistently slightly higher (Bonn: std 0.134 vs 0.116 for L2), making L2 the safer default.

C. Q3: Does Elastic Net Consistently Outperform L1/L2?

No — Elastic Net does not consistently outperform L1 or L2, and in one case severely underperforms both.

On UCI Pipeline A, Elastic Net achieves CV F1 = 0.186, compared to 0.437 (L1) and 0.448 (L2) — a 57% relative drop. With `l1_ratio=0.5`, the L1 penalty competes with L2 on only 8 correlated features, producing an unstable coefficient path that neither effectively selects features nor smoothly shrinks them. On MIT and Bonn Pipeline B, all three regularisers converge to near-identical performance because the 2 PCA components already minimise complexity — regularisation strategy becomes secondary.

The critical lesson: Elastic Net’s theoretical advantage only materialises when `l1_ratio` is tuned to the dataset’s correlation structure. Without grid search, it introduces complexity without benefit.

D. Q4: How Does Imbalance Handling Interact with Regularisation?

Imbalance handling has a larger effect on Recall than regularisation choice, and the two interact in a dataset-size-dependent manner.

On **small datasets** (Bonn, 400 samples), aggressive resampling is counterproductive. Undersampling leaves only ~ 128 training samples, destabilising any regulariser’s coefficient path. SMOTE risks generating synthetic samples overlapping with the majority-class decision boundary. Class Weighting, which

modifies only the loss function, is the safest choice — it ties SMOTE at F1 = 0.857 and PR-AUC = 0.907 while operating on the full 320 training samples.

On **large datasets** (MIT, 11,500 samples), all imbalance techniques perform equivalently on Pipeline B (F1: 0.867–0.886), confirming that strong features reduce the practical impact of class imbalance.

On **medium datasets** (UCI Pipeline A), imbalance handling is the primary bottleneck: without any correction, recall collapses to 0.02 regardless of regulariser. With SMOTE or under-sampling, all regularisers achieve F1 ≈ 0.44 . The bottleneck was imbalance, not regularisation.

IX. DISCUSSION

This study demonstrates three principal findings with consistent cross-dataset evidence.

Finding 1: Feature representation quality dominates all other choices. The F1 gap between pipelines (up to +0.592 on MIT) far exceeds gaps attributable to regulariser choice (max ~ 0.03) or imbalance technique (max ~ 0.02). Investing in domain-appropriate feature engineering yields returns that no amount of regularisation tuning can replicate from poor features.

Finding 2: Dataset size determines which imbalance technique to use. On large datasets (MIT), all techniques perform equivalently. On small datasets (Bonn), Class Weighting is preferable as it does not reduce training data or introduce synthetic samples in sparse feature spaces.

Finding 3: Simpler regularisers generalise better when features are pre-compressed. When Pipeline B reduces data to 2 PCA components, regularisation choice is nearly irrelevant. Careful feature engineering is a substitute for complex regularisation — they are not complementary in this setting.

X. CONCLUSION

This study systematically evaluated preprocessing, regularisation, and imbalance handling for epileptic seizure prediction across three heterogeneous EEG datasets. Key findings:

- 1. Preprocessing order critically affects generalisation.** Pipeline B outperformed Pipeline A by up to F1 = +0.592 (MIT). The effect scales with feature dimensionality and is negligible when features are already low-dimensional (UCI).
- 2. L2 regularisation generalises most stably** across all three datasets. L1 is competitive and provides meaningful sparsity (30% on Bonn); both outperform Elastic Net without hyperparameter tuning.
- 3. Elastic Net does not consistently outperform L1/L2.** With default `l1_ratio=0.5`, it severely underperforms on UCI (F1 = 0.186 vs 0.448). Grid search over `l1_ratio` is mandatory.
- 4. Imbalance handling effect is dataset-size dependent.** On large datasets (MIT), all techniques are equivalent. On small datasets (Bonn), Class Weighting is preferred.

On medium imbalanced data (UCI), all techniques yield $\sim 30\times$ recall improvement over no handling.

5. **Best overall combinations:** UCI: Pipeline A + L2 + any imbalance technique; MIT: Pipeline B + L2 + Class Weighting (F1 = 0.886, PR-AUC = 0.955); Bonn: Pipeline B + L1 or L2 + Class Weighting (F1 = 0.857, PR-AUC = 0.907).

Future work: non-linear models (Random Forest, SVM-RBF) evaluated on Pipeline B features; grid search over Elastic Net `l1_ratio`; and patient-specific cross-validation for clinical deployment assessment.